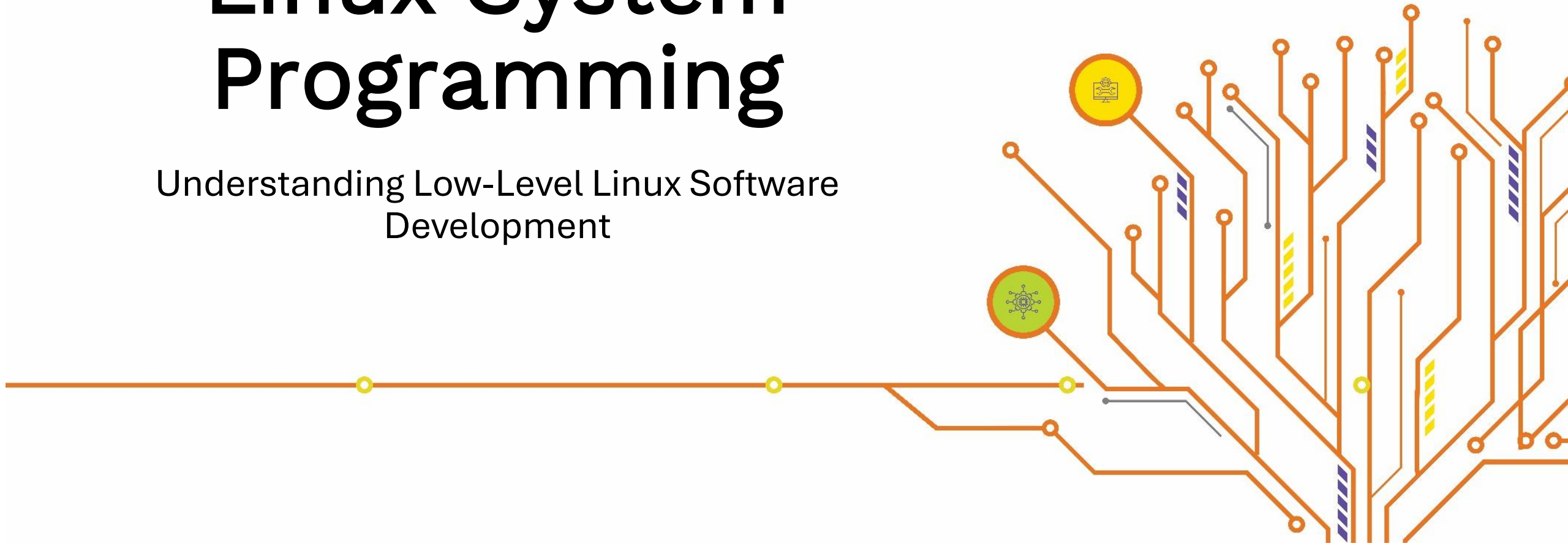
KPIT | APEX LAB

Introduction to Linux System Programming

Understanding Low-Level Linux Software Development



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What is System Software?

system software is the foundational layer of software

Core software that manages hardware and provides fundamental services.

- enables the operating system, hardware, and core execution environment to function
- provides the essential services upon which application software depends
- typically includes components that interface closely with the kernel and C library and provide low-level services to applications

Application Software
GUI, scripts, tools

System Software
OS Kernel, Drivers, Libraries

Hardware
CPU, Memory, IO

Example of System Software

Examples

- **GUI frameworks / window systems -** X11, Wayland, Weston, GTK core services
- **Compiler toolchain -** GCC/Clang, assembler, linker (binutils)
- **Debugger -** GDB, LLDB
- **System daemons -** init systems, udev, dbus
- **Web server-** Nginx, Apache (provides infrastructure services to other apps)
- **Database engines -** MySQL, SQLite, PostgreSQL (foundational services used by applications)

System Software Vs Application Software

System software:

- Avoids large frameworks (Qt, Java, Node.js, .NET)
- Uses low-level APIs (syscalls, POSIX, libc)
- Often deals with raw data, hardware access, and OS internals
- Example: Writing a device access program using open()

Application software:

- Uses GUI toolkits, web frameworks, networking libraries
- Relies on abstracted APIs and toolkits that hide OS details
- Example: Python's requests library

System Software Does NOT Use Higher-Level Libraries

System Software Works With Fewer Abstractions

System software must understand:

- Memory layout
- Processes, threads, scheduling
- Filesystem internals
- Drivers and hardware behavior
- Networking stack primitives

Application path → high-level libraries → kernel

Applications, by contrast, get simplified abstractions:

- High-level I/O
- Framework-provided networking
- GUI abstraction layers
- Scripting languages

System program path → libc → syscalls → kernel

System Programming

- System programming is the discipline used to create system software.
- Writing software that interacts with system software and the OS kernel.
- Uses system calls to control processes, memory, I/O, and devices.
- Typically written in C/C++
- Close to hardware, performance-critical, and resource-aware.

System Programming → Creates System Software → Supports Application Software

System Programming

- System software interfaces with the kernel, libc, and low-level OS mechanisms, its development requires **system programming skills** such as:
  - Writing C programs that use POSIX APIs and system calls
  - Understanding Linux kernel behaviour
  - Managing memory manually
  - Handling low-level concurrency (threads, mutexes, semaphores)
  - Interacting with hardware through drivers or /dev

Why System Programming?

Performance: direct, efficient control over CPU, memory, I/O.

Determinism: required for embedded, automotive, real-time workloads.

Hardware access: reading sensors, controlling motors, talking to peripherals.

Creating foundational software: kernels, drivers, daemons, system libraries.

Better debugging and root-cause analysis: understand how the OS behaves.

automotive ECUs require deterministic behavior → system programming essential

Key Areas of System Programming

- Process creation and management
- File and device I/O
- Multithreading & synchronization
- Memory allocation & mapping
- Interprocess communication (IPC)
- Networking (sockets)
- Signals & timers
- Error handling and recovery

Cornerstones of System Programming

System Calls (syscalls)

C Library (libc)

C Compiler/linker (toolchain)

App → libc → syscall → kernel

libc translates C API → syscall ABI

System Calls

- System software interacts directly with the Linux kernel through:
  - **System calls** (ex: open(), read(), write())
  - **Device interfaces** (/dev, sysfs, procfs)
  - **Process, memory, and file operations**
- This layer understands how to communicate with kernel primitives rather than relying on high-level abstractions.

C Library (libc)

System software relies heavily on the **standard C library** (glibc, musl, uClibc), which provides:

- System call wrappers
- Memory allocation (malloc, free)
- Process control (fork, exec)
- File I/O (read, write)
- The C library forms the API boundary between user space and the kernel.

libc implements POSIX API, not all system calls

API — Application Programming Interface

“What the programmer sees.”

- An **API** is the *interface you program against*.

Examples

- POSIX API (open(), read(), write(), fork())
- C library functions (printf(), malloc())
- Driver user-space interfaces (ioctl, /sys, /proc)
- Linux system call interface *as declared in user-space headers*

ABI — Application Binary Interface

- An **ABI** describes *how compiled programs interact at the binary level*.
- It covers how **functions, data structures, calling conventions, and binaries** are laid out in memory.

“What the machine sees.”

Examples of ABI Elements

- How function parameters are passed (registers/stack)
- Binary format (ELF)
- Instruction set architecture (ARM, ARM64, x86)
- Symbol names (mangling)
- Struct memory layout (alignment, padding)
- System call numbers
- Shared library versioning (libc.so.6)

API vs ABI

Aspect	API	ABI
Visible to	Programmer	Machine / binary loader
Defined by	Headers, documentation	Compiler, linker, architecture
Enforced at	Compile time	Runtime
Stability needed for	Rebuilding code	Running already-compiled binaries
Breakage effect	Needs recompilation	Existing binaries break
Examples	POSIX, glibc functions	ELF format, calling conventions, system call numbers

Why API & ABI Matter?

- Cross-compiling
- Different CPU architectures = different ABIs.
- Shared libraries
- You can update libc, libm, or driver libraries, and all apps continue working — **only because ABI compatibility is maintained**.
- Kernel–User Space Interaction
- Syscalls are part of the **kernel ABI**
- Compiler / Toolchain
- The toolchain is built around a specific ABI:
- GCC configuration (--with-abi, --with-arch)
- glibc configuration
- Linker, assembler rules

POSIX - Portable Operating System Interface

- C APIs for Unix-like operating systems
- IEEE Std 1003.1-1990
- ISO/IEC 9945-1:1990
- POSIX defines **C code API** for many concepts
  - File and directory operations
  - Clocks and timers
  - Semaphores
  - Shared memory
  - Threads

- Official list of headers: http://pubs.opengroup.org/onlinepubs/9699919 799/idx/head.html
- Some familiar ones: <stdio.h>, <stdint.h>

Summary

- System programming = low-level control
- Uses syscalls + libc
- API vs ABI distinction
- POSIX standard